



TRANSDUCER INERTIA AND STINGER STIFFNESS EFFECTS ON FRF MEASUREMENTS[†]

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In order to study methods for measuring complete driving point frequency response function (FRF) characteristics for forces and moments, it was necessary to evaluate which test technique had the best chances of success. Impulse testing has consistency problems with input magnitudes and direction as well as mixing windows for signal processing. The chirp has excellent signal-to-noise ratios and generally superior coherence when compared to both random and impulse testing. It was found that stinger resonances within the test frequency range and later outside the test frequency range caused measurement complications for certain test structure natural frequencies. These complications were effectively predicted using the transducer's inertial properties, the stinger's stiffness properties, and the structure's linear and angular driving point FRF characteristics (as determined from impulse tests) when stinger resonance effects were removed from the chirp data.

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1. INTRODUCTION

The test set-up has significant effects on the measured frequency response functions (FRFs). One of the main contributors is the stinger as reported by Mitchell and Elliot [1], Hieber [2], Hu [3], Chiang and Chou [4], and McConnell *et al.* [5]. A second factor of concern is the effect of mass on the measured FRF results as discussed by Ewins [6], McConnell [7], Ashory and Ewins [8] and Ashory [9]. The importance of these various effects depends on what we are using the FRFs for. If we need only approximate mode shapes and natural frequencies, we can often ignore these effects without significant consequences. Consider the situation where we are concerned with testing two or more substructures with the aim of connecting these substructures together at multiple points into a larger structure as described by Varoto [10]. Then, we want to predict the dynamic behaviour of the assembled structure based on the substructure measurements. Sestieri *et al.* [11] have shown that the testing requirements are significantly different, particularly when dealing with light-weight structures as they did. In Sestieri's [11] case, both linear and angular measurements were needed.

The main task of the present paper is to evaluate transducer interaction with a simple structure when it is excited by different excitation time histories such as impulse, chirp and random. The evaluation will be done on data from a free-free beam.

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2. BASIC BACKGROUND FORMULATION

In order to grasp the situation more fully, consider the input-output relationships between two points, called p and q , of a given structure under test (SUT). The input at q consists of both a force $\mathbf{F}$ and a moment $\mathbf{M}$ where these two inputs are vectors, each having three scalar components. The output at p consists of both a linear acceleration $\mathbf{a}$ and an angular acceleration α where these two outputs are vectors, each having three scalar components. These six input and output variables are frequency dependent so that there are 36 frequency-dependent input-output relationships that can be written as

$$\begin{Bmatrix} a \\ \alpha \end{Bmatrix} = \begin{bmatrix} A(\omega) & B(\omega) \\ C(\omega) & D(\omega) \end{bmatrix} \begin{Bmatrix} F \\ M \end{Bmatrix} = [H(\omega)] \begin{Bmatrix} F \\ M \end{Bmatrix}. \quad (1)$$

The overall accelerance matrix $H(\omega)$ is symmetrical about its diagonal. This 6×6 matrix can be conveniently subdivided into four 3×3 sub-matrices called A , B , C and D . The diagonal sub-matrices A and D are symmetrical while the off-diagonal sub-matrices B and C are transposes of one another; i.e. $B = C^T$ and/or $C = B^T$. In the special one-dimensional case considered here, there is only a single linear acceleration and a single angular acceleration so that $B = C$. Now, let us examine the consequences of attaching a rigid structure to the SUT at location p . The rigid structure has inertial properties of mass m_p and mass moment of inertia J_p . The structure is assumed to have a single linear acceleration a_p and a single angular acceleration α_p while the inputs are a single force F_q with zero moment M_q . For this case, Equation (1) reduces to

$$\begin{bmatrix} 1 + m_p A_{pq} & J_p B_{pq} \\ m_p C_{pq} & 1 + J_p D_{pq} \end{bmatrix} \begin{Bmatrix} a_p \\ \alpha_p \end{Bmatrix} = \begin{Bmatrix} A_{pq} \\ C_{pq} \end{Bmatrix} F_q \quad (2)$$

$$\begin{Bmatrix} a_p \\ \alpha_p \end{Bmatrix} \begin{bmatrix} 1 + m_p A_{pq} & J_p B_{pq} \\ m_p C_{pq} & 1 + J_p D_{pq} \end{bmatrix}^{-1} \begin{Bmatrix} A_{pq} \\ C_{pq} \end{Bmatrix} F_q = \begin{Bmatrix} A_{pq}^m \\ C_{pq}^m \end{Bmatrix} F_q \quad (3)$$

$$\begin{Bmatrix} A_{pq}^m \\ C_{pq}^m \end{Bmatrix} = \begin{bmatrix} 1 + m_p A_{pq} & J_p B_{pq} \\ m_p C_{pq} & 1 + J_p D_{pq} \end{bmatrix}^{-1} \begin{Bmatrix} A_{pq} \\ C_{pq} \end{Bmatrix}. \quad (4)$$

It is clear from equation (4) that the measured linear accelerance A_{pq}^m and angular accelerance C_{pq}^m are contaminated not only by the mass m_p and its mass moment of inertia J_p , but also by the four accelerances A_{pq} , B_{pq} , C_{pq} and D_{pq} . This means that Ashory's [9] results are not always correct depending on the relative values of accelerances A , B , C and D as well as on the inertias m_p and J_p . In addition, we see that the measured accelerances are the actual accelerances when m_p and J_p are zero. Thus, it appears that attaching transducers and/or stingers to a light-weight structure can lead to contamination of the measured results.

For example, consider the two points 1 and 2 as shown in Fig. 1 for the free-free beam. Two different test points of view are now examined. In the first point of view, the excitation force F_1 is applied at location 1 while $M_1 = 0$. In this case, we can measure the resulting accelerations a_1 , a_2 , and α_1 which are given by

$$a_1 = A_{11} F_1 \quad (5)$$

$$a_2 = A_{21} F_1 \quad (6)$$

$$\alpha_1 = C_{11} F_1. \quad (7)$$

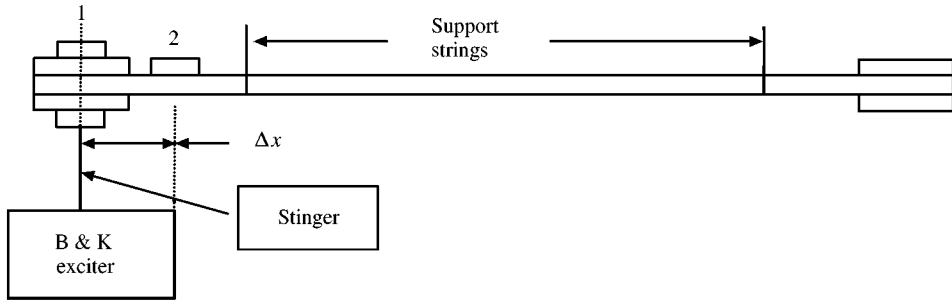


Figure 1. General test set-up of free-free beam for either impulse- or exciter-driven testing with two miniature accelerometers, one at each of points 1 and 2.

From equations (5)–(7), we see that we can directly measure the FRFs A_{11} , A_{21} and C_{11} if we can measure the angular acceleration α_1 . If we cannot measure α_1 , then we need to use the local rigid-body approximation that $\alpha_1 = (a_2 - a_1)/\Delta x$ from which we obtain C_{11} as

$$C_{11} = \frac{A_{21} - A_{11}}{\Delta x} \quad (8)$$

when α_1 is positive counterclockwise. Equation (8) represents an indirect method for determining the angular acceleration FRF C_{11} from the two linear acceleration FRFs A_{21} and A_{11} . These two linear acceleration FRFs are obtained when the excitation is applied at location 1.

In the second point of view, the structure is excited at location 2 with a force F_2 while the moment $M_2 = 0$, and we assume that we can measure a_1 , a_2 and α_1 as before. However, we can also think of the loading at point 2 in a local rigid body sense as being equivalent to a force and couple or torque being applied at location 1 so that

$$M_1 = \Delta x F_2 \quad (9)$$

$$F_1 = F_2$$

The corresponding outputs become

$$a_1 = A_{12}F_2 = A_{11}F_2 + B_{11}\Delta x F_2 \quad (10)$$

$$a_2 = A_{22}F_2 \quad (11)$$

Equation (11) shows that A_{22} can be measured directly. In this case, the angular acceleration becomes

$$\alpha_1 = C_{11}F_2 + D_{11}\Delta x F_2 \approx \frac{a_2 - a_1}{\Delta x}. \quad (12)$$

From equation (10) we can solve for linear acceleration due to moments, B_{11} , to obtain

$$B_{11} = \frac{A_{12} - A_{11}}{\Delta x}. \quad (13)$$

From equation (13), it is clear that: (a) we are mixing FRFs obtained from excitation at both points 1 and 2; (b) we are subtracting two potentially large numbers, and, finally (c) B_{11} has basically the same definition as C_{11} since ideally $A_{12} = A_{21}$ in this one-dimensional case.

From equation (12) we can obtain the angular accelerance due to moments, D_{11} , as

$$D_{11} = \frac{A_{22} - A_{12}}{(\Delta x)^2} - \frac{C_{11}}{\Delta x} = \frac{(A_{22} + A_{11}) - (A_{12} + A_{21})}{(\Delta x)^2}. \quad (14)$$

From equation (14) it is clear that: (a) we are mixing FRFs obtained from excitation at two different points (1 and 2 in this case), and (b) two numbers are being added together in each bracket, and then subtracted from one another so that noise can be a significant factor.

In either case, the use of a subtraction scheme can easily lead to large errors in the estimate of either B_{11} , C_{11} or D_{11} . Thus, it is desirable to measure α_1 directly with an angular accelerometer so that C_{11} can be obtained directly as shown in equation (7). Then, B_{11} is also immediately known since $B_{11} = C_{11}$ in this case. The only remaining term with high uncertainty is D_{11} although a direct measure of C_{11} should give better results than using the difference form on the right-hand side of equation (14).

3. EXPERIMENTAL ANALYSIS

3.1. EVALUATION OF EXCITATION CHOICES

Consider the test set-up shown in Fig. 1 where a free-free steel beam $6.35 \times 25.4 \times 609.6$ mm is supported by light strings attached at the quarter points. Two small steel plates $25.4 \times 25.4 \times 6.35$ mm are welded to each end. Point 1 is the middle of the two steel plates at the left end and point 2 is located 38.1 mm to the right as shown. The excitation at either location 1 or 2 is provided by either a PCB model 086C03 impulse hammer or a Bruel & Kjaer Model 4809 vibration exciter driven by a Bruel & Kjaer Model 2706 power amplifier. The stinger force is measured using a PCB Model 288C01 Impedance Head. The brass stinger is 3.0 mm in diameter and 127 mm long. The acceleration is measured using two miniature PCB Model 353B16 accelerometers. A Kistler TAP linear/angular accelerometer was damaged while setting up one of the early tests so all results are obtained by using the two miniature accelerometers. Due to the fact that the B, C, and D FRFs are determined by using the difference in the accelerance FRFs, as shown in the previous paragraph, the two miniature accelerometers are required to be calibrated to have matched outputs. These accelerometers were mounted on the same rigid block and the sensitivity of one accelerometer was adjusted until the mean value of the transmissibility ratio between the two accelerometers was unity over the frequency range of interest.

The beam was excited using three different excitation schemes in the 0–625 Hz frequency range. First, impulse tests were conducted where the structure was impacted by the PCB Model 086C03 impulse hammer. The PCB miniature accelerometers were used at locations 1 and 2. The data were processed using a Data Physics DP420 three-channel digital signal analyser. The impulse force signal was analysed using a standard rectangular window while the impulse response used an exponential window with 5% window leakage. The resulting impulse FRFs are based on the average of 6 data sets.

The vibration exciter was used as shown in Fig. 1 for the chirp and random excitation cases. For the chirp excitation, rectangular windows were used for both the input and output signals. The chirp started at 5% of the window width and continued to 80% of the window. The response vibration ceased before the end of the window due to the high electrodynamic damping of the exciter when the input voltage goes to zero. The resulting chirp FRFs contain the average of 20 data sets. The random excitation used Hanning data windows and an average of 100 data sets in calculating the FRFs. The input force root-mean square (rms) level was approximately 0.25 N rms for both the chirp and random data so that the input power levels are comparable on an rms basis.

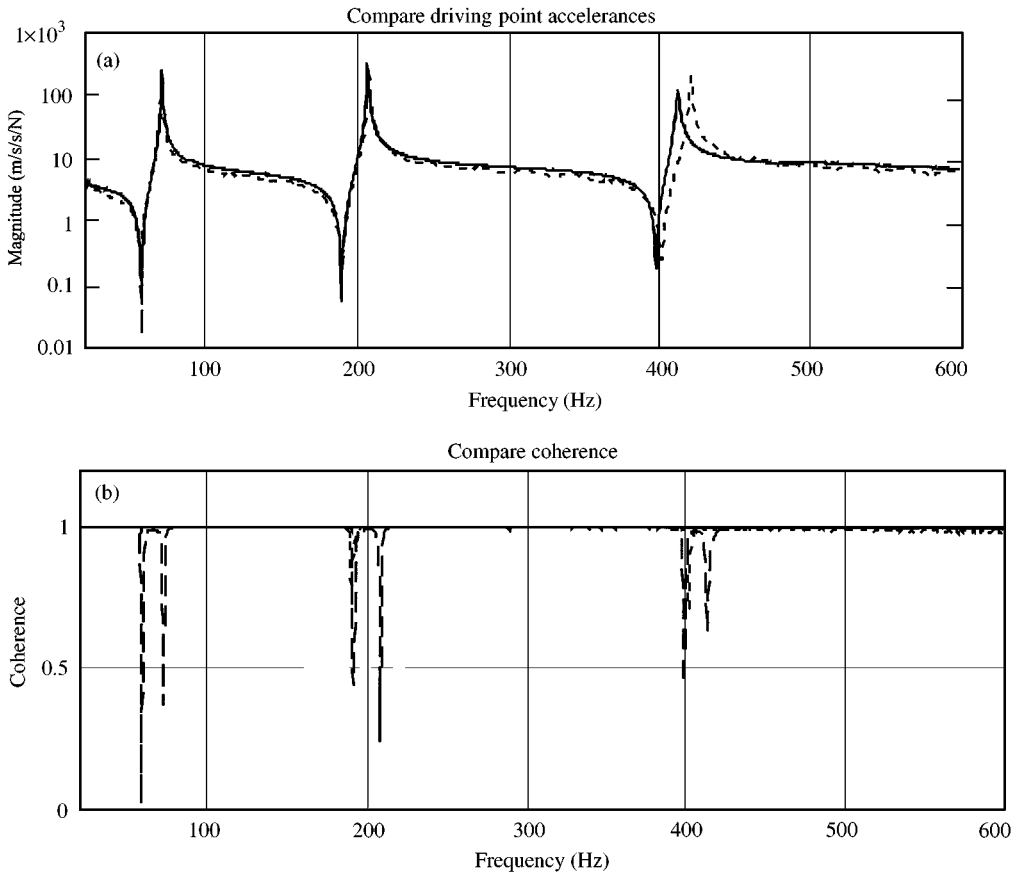


Figure 2. Comparison of A_{11} FRFs (a) and coherence (b) for chirp, impulse, and random excitation cases: (—) chirp, (---) impulse, (- -) random.

Figure 2(a) shows the driving point FRF A_{11} with units of $\text{m/s}^2/\text{N}$ plotted against frequency in Hz. Both the low-frequency beam suspension and the exciter/stinger natural frequencies are below 20 Hz [4]. An examination of this figure indicates that the three antiresonance frequencies and the first two natural frequencies are in good agreement. However, the third resonance peak region presents some disagreement in the antiresonance frequency, the natural frequency and the magnitudes dependent on the excitation method employed. The chirp and random excitation methods appear to be in close agreement compared to the impulse excited data. Figure 2(b) shows a comparison of the coherence function for the three FRFs in Fig. 2(a). It can be observed that the chirp excitation has a coherence close to unity at all frequencies. The impulse excitation has a good coherence close to unity except in the antiresonance region where the accelerometer noise floor represents a problem. The random results have poor coherence at both the antiresonance and the resonance regions. The explanation for this poor coherence at the antiresonance frequencies is the same as for the chirp tests, i.e. low acceleration signals in the presence of instrument floor noise. The reason the random excited FRF resonances have poor coherence is that the force input signal is very low due to the exciter force drop-out that occurs around SUT resonances.

The only reasonable explanation for the lowered third natural frequency from the chirp and random excited test results compared to the impulse test results is that the test set-up violates the free-free boundary condition and introduces some extraneous force and/or moments that are not measured. However, this force/moment must be in or out-of-phase with the measured force so that the output is coherent with the input in the chirp excitation case. These results are in agreement with the predictions of McConnell *et al.* [5] where stinger dynamics are shown to have the capability to play an important role in the test results.

Some conclusions can now be drawn on the excitation methods. (1) Careful impulse tests meet the boundary condition requirements and, generally, have near unity coherence everywhere except at the antiresonance regions. However, impulse tests give distorted damping estimates due to the use of exponential windows on the output acceleration. Hence, the impulse test FRF has peaks that are below the other curves between resonance and antiresonance regions; and finally, has higher notches (are not as deep) compared to the chirp results. (2) The chirp test has the best coherence overall but has a troublesome shift in the third natural frequency due, in this case, to the test set-up. (3) The random excitation has the poorest coherence. Part of this poor coherence is due to the fact that most of the data occur in the lower quarter of the A/D converter's range. Thus, for the same input rms force level, the A/D converter's full scale must be in excess of four to five times the rms level in order to prevent signal clipping. In the chirp case, the full-scale A/D converter level needs to only exceed the rms level by about 40%. Consequently, the signal-to-noise ratio is a less significant factor for the chirp test compared to the random test. Finally, only the impulse and chirp test method results are used in the rest of this paper since they give the best data from each test method, i.e. impulse *vs* exciter.

3.2. FINDING THE A , B , C AND D MATRICES FROM IMPULSE AND CHIRP DATA

In this section the resulting values of the driving point accelerances A_{11} , B_{11} , C_{11} and D_{11} when the excitation is either a chirp or an impulse are examined. In Fig. 3, the directly measured linear/force accelerance A_{11} is compared for the impulse and chirp excitations. The first two resonant and antiresonant frequencies are in good agreement. However, the impulse FRF is lower than the chirp FRF due to the extra damping of the exponential window that must be used to prevent serious window leakage.

There is a significant difference in the third natural frequency where the measured chirp test estimate is 410 Hz while the measured impulse test estimate is 420 Hz. Also, it is observed that there is a small difference in the antiresonance frequencies. In addition, there is a significant difference in the peak values where the impulse peak is larger than the chirp peak, a result that is in contrast with the first two peaks. Some mechanism exists by which energy is being removed from the test beam that is not being measured. This mechanism is believed to involve flexural vibrations of the stinger.

The driving point linear accelerance due to moments, B_{11} , is shown in Fig. 4 as determined from equation (13). It is clear that the chirp data are sharper around the antiresonance regions and have higher peaks due to the better quality of data as indicated by the good coherence values. The noise in the impulse data near the antiresonance regions is emphasized when the data are subtracted. The double peaks in the chirp data at the third natural frequency require additional comments. From equation (13) we see that we are subtracting two quantities ($A_{12} - A_{11}$) where the force transducer and stinger are attached at point 1 for the A_{11} measurement and at point 2 for the A_{12} measurement. A subsequent independent set of measurements showed that the third natural frequency changed depending on where the force transducer and stinger were attached to the SUT. The driving point

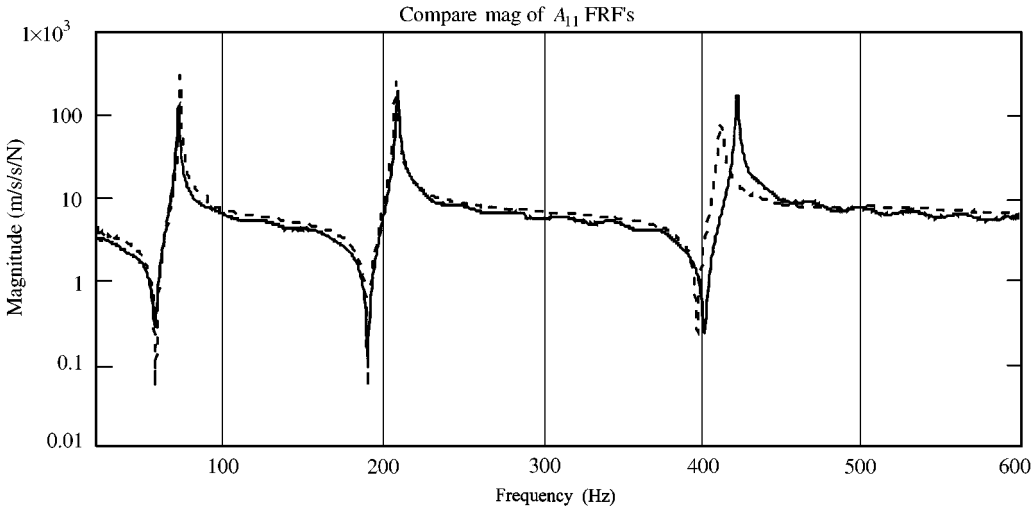


Figure 3. Comparison of linear driving point acceleration FRF A_{11} from impulse and chirp excitation: (—) impact, (---) chirp.

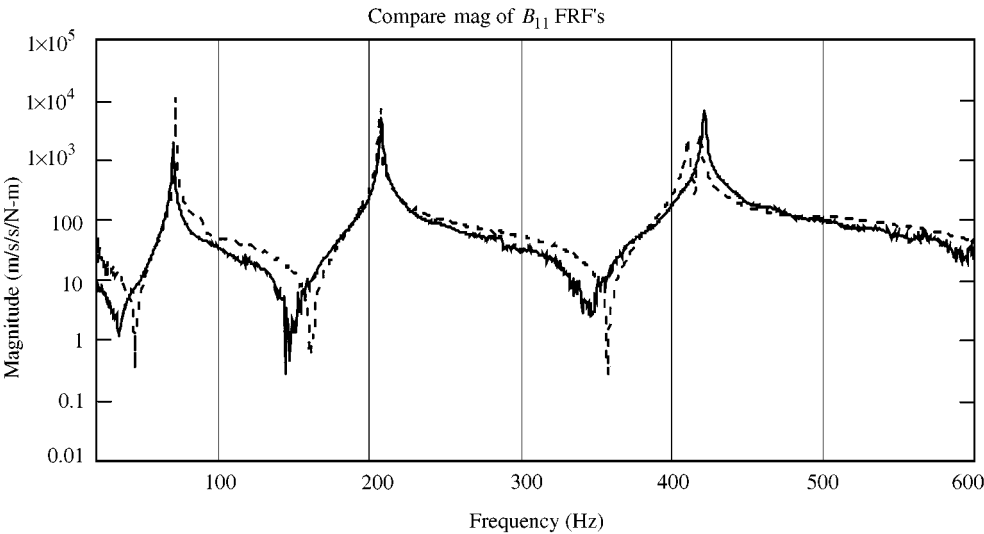


Figure 4. Comparison of linear driving point acceleration B_{11} from impulse and chirp excitation methods: (—) impact, (---) chirp.

accelerance C_{11} is calculated using equation (8) and is shown in Fig. 5. In this case, it is seen that the first two resonant frequencies are in good agreement and the impact peaks are lower than the chirp peaks. The antiresonance regions match much better in this case when compared with the B_{11} results. The impact valley values are seen to be higher and show more scatter. The third natural frequency has a single peak for the chirp excitation. An examination of equation (8) shows that the C_{11} difference is calculated only from point 1 excitation data, namely, $A_{21} - A_{11}$, so that the apparent resonance is the same for each data set.

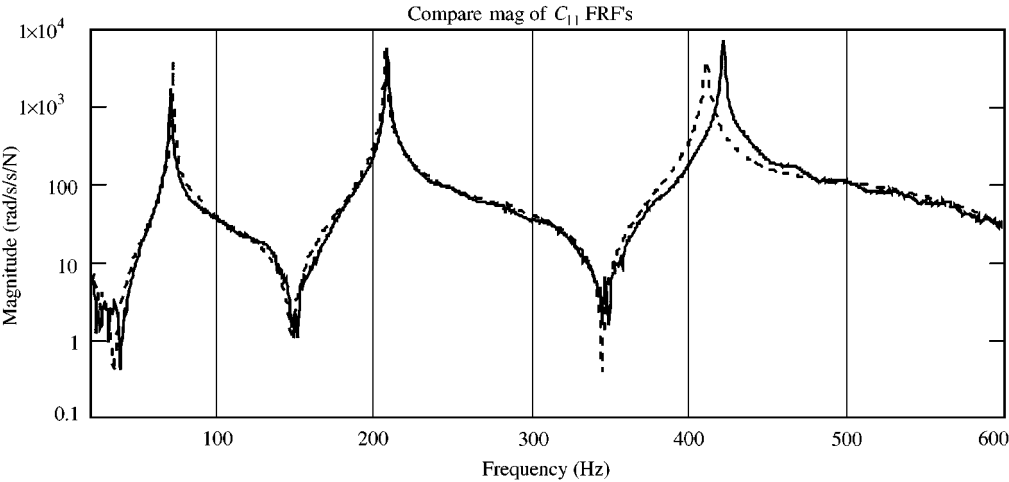


Figure 5. Comparison of linear driving point accelerance C_{11} from impulse and chirp excitation methods: (—) impact, (---) chirp.

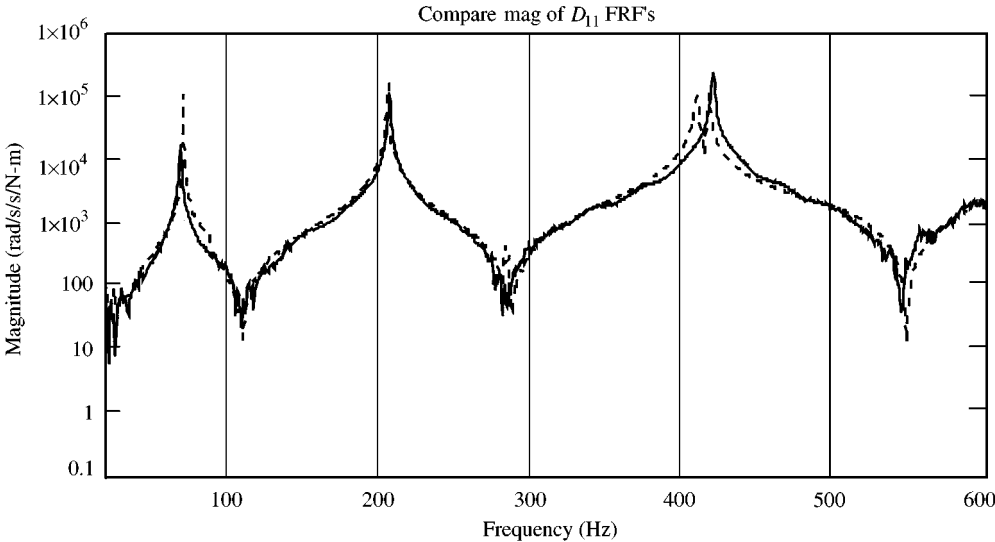


Figure 6. Comparison of angular driving point accelerance D_{11} from impulse and chirp excitation methods: (—) impact, (---) chirp.

The angular driving point accelerance D_{11} is calculated using equation (14) and the results are shown in Fig. 6. Again, there is good agreement for the first and second natural frequencies with the chirp data showing much sharper and deeper antiresonance with considerably less noise. The first two impact peaks are also lower than the chirp data peaks. However, the chirp data show two peaks in the vicinity of the third natural frequency. These double peaks are due to mixing data from excitation at points 1 and 2 as seen in equation (14). Consequently, this double-point excitation approach has the potential to be troublesome anytime we need to subtract FRFs.

3.3. STINGER AND INERTIA TEST SYSTEM EFFECTS

In this section, we shall apply equation (4) to the impact data and try to predict the behaviour obtained for the chirp data. The impedance head force transducer mass and mounting screw have an overall mass value of $m_p = 20.6 \times 10^{-3}$ kg and a mass moment of inertia value of $I_p = 1.2 \times 10^{-5}$ kg m². The stinger stiffness, considered as a simple spring, is $K = 15.1$ N m/rad. In equation (4), we can replace the mass moment of inertia J_p by $(I_p - K/\omega^2)$. Then, we can use equation (4) to predict the chirp measured values A_{11}^m and C_{11}^m , if we assume that the measured impulse data for A_{11} , B_{11} , C_{11} and D_{11} are correct. The predicted A_{11}^m FRF is shown and compared with the measured impulse and chirp data in Fig. 7 indicating that the first two peaks are in close agreement. The third peak shows that the corrected peak shift lies nearly midway between the other two peaks. The corresponding angular accelerance C_{11}^m correction results are shown in Fig. 8. These results are nearly identical to those obtained for A_{11}^m as shown in Fig. 7.

The measured peaks are 410.7 Hz from the chirp test and 420.3 Hz from the impulse test. The correction scheme of equation (8) predicts a peaks at 416.5 Hz when the effects of transducer mass and mass moment of inertia are included. The measured peaks are 410.7 Hz

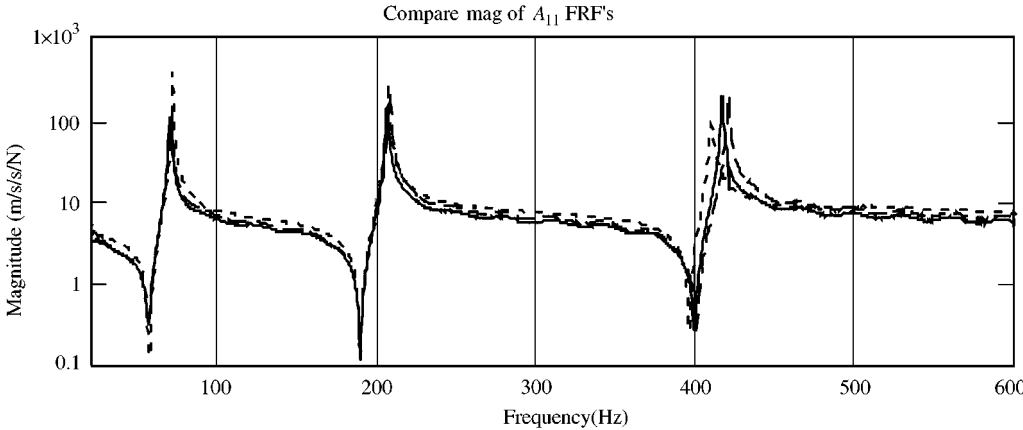


Figure 7. Comparison of measured impact and chirp linear driving point accelerance A_{11} data with the corrected impact data to simulate the chirp data: (—) corrected, (---) chirp, (···) impact.

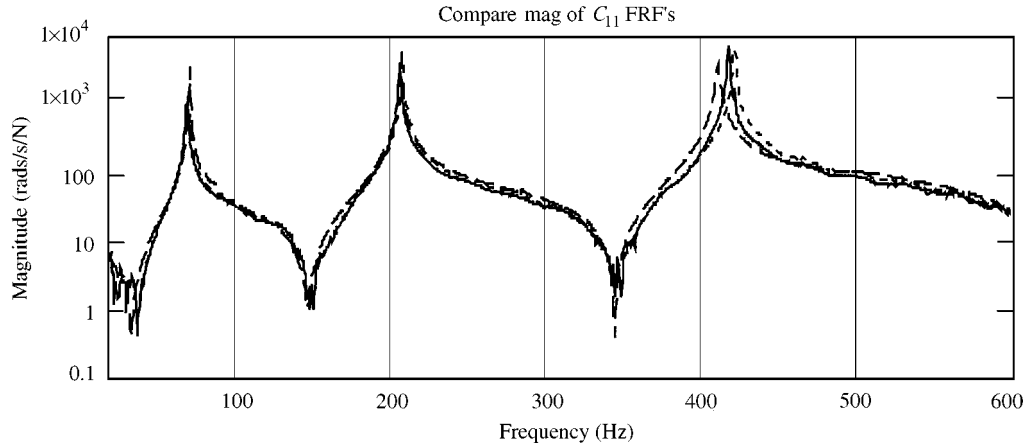


Figure 8. Comparison of measured impact and chirp linear driving point accelerance C_{11} data with the corrected impact data to simulate the chirp data: (—) corrected, (---) impact, (···) chirp.

from the chirp test and 420.3 Hz from the impulse test. The correction scheme of equation (4) predicts a peak at 416.5 Hz when the effects induced by the transducer mass and its moment of inertia are included. When the stinger is removed and the accelerance impedance head is attached at location 1 and the beam is impacted by the hammer, the measured third natural frequency is equal to 417 Hz. Hence, we conclude that the correction scheme of equation (4) works by accounting for the frequency shift due to the force transducer's mass and mass moment of inertia. The problem is to account for the additional approximately 6 Hz drop in frequency from 417 to about 411 Hz. Subsequent tests revealed that there is a stinger resonance at about 675 Hz. This resonance is unduly influencing the third natural frequency measurements in a manner similar to that predicted by McConnell *et al.* [5]. The amount of frequency shift is significantly reduced to about 418 Hz when the arc of the stinger is constrained so that it cannot vibrate in the lateral direction.

4. REASONS FOR OBSERVED RESULTS

This study revealed several significant observations relative to coherence issues, hardware employed to excite and measure the data, test boundary conditions and methods of calculating the angular accelerance FRFs from the linear accelerance FRFs. We shall summarise the relevant ideas here and their implications to modal testing in general.

(1) *General theoretical observations* show that we need instruments to measure both angular acceleration and moments if the 36 potential input-output quantities are to be measured. It was found that, in principle, the angular accelerance can be calculated from the difference between two linear accelerances of two close points. The major difficulty with this method is that often two large quantities are being subtracted from one another so that noise is an inherent part of the problem.

(2) *Boundary conditions* were altered in two ways. First, the effect of transducer mass and mass moment inertia on the measured accelerances were modelled. Second, the effect of stinger stiffness was also modelled. The transducer inertia caused a minor frequency shift in this case. However, the transducer inertia and stinger stiffness combined to have an out-of-range natural frequency which significantly influenced the measured results.

(3) *Calculation of angular accelerance FRFs from two linear accelerance FRF measurements* appears to be straightforward from a theoretical point of view. However, in practice, if the accelerances are obtained from a single driving point as shown in equation (8), a single third resonance occurs. On the other hand, if two accelerances were used that have different driving points as given in equation (13), a double peak occurs for the third resonance due to shifts in natural frequency owing to stinger and inertial effects. In addition, the angular accelerance is obtained by subtracting one accelerance from another. This calculation method leads to serious noise problems when the two values are very close.

(4) *Coherence* issues were present in the random and impact test methods and not significant in the chirp test.

Random signal tests: Random signal tests produced the poorest coherence at resonance peaks and antiresonant notches and this is due to several factors that can affect all random tests. First, the A/D converter amplifier gain must be adjusted to handle peak values that are at approximately 4–5 times the signal's rms level. Hence, we are working with only the bottom 20–25% of the A/D converter's dynamic range.

Second, the electrodynamic vibration exciter has significant force-drop out at test system resonance due to its back emf characteristics and the large structural motion at resonance. This means that the input force becomes very small compared to that available at other

frequencies so this signal is comprised in terms of both measurement system noise and A/D converter lack of resolution. Hence, the poor coherence at resonance. Third, the acceleration signal becomes small at antiresonance so that this signal becomes susceptible to measurement system threshold noise and lack of A/D converter range as well. Hence, the poor coherence at antiresonance notches.

Impact tests: Low coherence occurs in this case due to low signal levels due to instrument threshold noise and the use of the lower end of the A/D converter's dynamic range. Generally, the coherence is lower for driving point measurements compared to transfer measurements. This is due to the fact that driving point acceleration signals have a very large peak value during the impact with much smaller values in the ensuing transient vibration. This large signal causes the A/D converter to be set so that we are working with the lower end of the converter's dynamic range. In transfer acceleration measurements, the peak values are much smaller, and hence, more of the A/D converter's dynamic range can be used.

Chirp tests: This test has the best coherence for a number of reasons. First, the A/D converter's dynamic range can be used in a near-optimum manner. Second, there is no significant exciter force drop-out since there is insufficient time to build up significant response at each response. Third, we can use rectangular window functions if the analyser and the chirp signal length are properly adjusted for the SUT. Finally, the chirp gave essentially unity coherence even when there were measurement problems around the 3rd natural frequency due to the stinger's own response. This is a classic case where high coherence does not mean the data are good since the error mechanisms were repeatable relative to the true data and cannot be detected through coherence concepts.

5. CONCLUSIONS

This study examines the basic issues that are involved in attempting to measure the 2 degree-of-freedom (one linear and one angular) accelerances due to a force and a moment. It was found that transducer inertia (both mass and mass moment of inertia) and stinger stiffness as well as its own natural frequencies can play an important role in the measured results. These measured results are to be used in combining substructures with one another in the manner described by Varoto [10] and Sestieri *et al.* [11]. At the present time, it is apparent that a better test methodology is required to achieve the quality of data that is necessary to achieve a successful combination of substructure characteristics in order to predict couple behaviour.

The effects of signal choice to drive a structure under test as well as the structure transducer interaction were examined. These effects were observed in the driving point and transfer accelerance measurements. These measurements were obtained from two closely space monoaxial accelerometers when the structure is excited at one of these same two points.

Three excitation signals were used. The random test showed poor coherence at both resonant peaks and antiresonant valleys. The peaks had poor coherence due to exciter force drop-out at resonance where the measured input force is a small signal compared to the instruments full-scale range. The poor coherence at notches is due to the low acceleration signals compared to the full-scale range required to handle the large random peaks.

The chirp test showed the best overall coherence since there is insufficient time for the exciter force drop-out to occur since the chirp passes through the resonant frequency too fast for significant resonant response to occur. In addition, the input amplifier gains can be adjusted so that the full dynamic range of the A/D converters is used.

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APPENDIX: NOMENCLATURE

A	3×3 linear accelerance matrix due to force input F
A_{pq}	linear accelerance at location p due to force input at location q
A_{pq}^m	measured linear accelerance at location p due to force input at location q
a	linear acceleration vector
a_p	Acceleration at p th location
m_p	mass attached at p th location
B	3×3 linear accelerance matrix due to moment input M
B_{pq}	linear accelerance at location p due to moment input at location q
B_{pq}^m	measured angular accelerance at location p due to force input at location q
C	3×3 angular accelerance matrix due to force input F
C_{pq}	angular accelerance at location p due to force input at location q
C_{pq}^m	measured angular accelerance at location p due to force input at location q
D	3×3 angular accelerance matrix due to moment input M
D_{pq}	angular accelerance at location p due to moment input at location q
F	resultant force vector applied to SUT
F_q	single force applied to SUT at location q
J_p	mass moment of inertia attached to the SUT at location p
M	resultant moment vector applied to SUT
M_q	single moment applied to SUT at location q
FRF	frequency response function
SUT	structure under test